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SHEET CONVEYING DEVICE, IMAGE FORMING APPARATUS, AND SHEET PROCESSING APPARATUS

Abstract

A sheet conveying device includes a body, a first guide in the body, a cover, a second guide in the cover, a detector, and a positioner. The first and second guides form a conveyance path at a closed state of the cover. The detector detects a sheet in the conveyance path and includes a photosensor in the body and a feeler in the cover. The feeler includes a detected portion movable in conjunction with the sheet passing through the conveyance path and a contacted portion. The photosensor has a detection region in which the detected portion entering the detection region is detected. The positioner in the body contacts the contacted portion to position the detected portion in the detection region.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This patent application is based on and claims priority pursuant to 35 U.S.C. § 119 to Japanese Patent Application No. 2024-036040, filed on Mar. 8, 2024, in the Japan Patent Office, the entire disclosure of which is hereby incorporated by reference herein.

BACKGROUND

Technical Field

[0002] The present disclosure relates to a sheet conveying device to convey a sheet, an image forming apparatus incorporating the conveying device, and a sheet processing apparatus incorporating the conveying device. Such an image forming apparatus is, for example, a copier, a printer, a facsimile machine, or a multifunction peripheral having at least two of copying, printing, scanning, facsimile, and plotter functions.

Related Art

[0003] An image forming apparatus such as a copier or a printer known in the art includes a sheet conveying device. The sheet conveying device includes a cover that can be opened and closed to expose a part of or an entire of a sheet conveyance path. The cover is opened to remove a sheet jammed in the sheet conveyance path inside the body of the image forming apparatus.

SUMMARY

[0004] The present disclosure described herein provides a sheet conveying device including a body, a first guide, a cover, a second guide, a detector, a feeler, and a positioner. The first guide is in the body. The cover is openably closable the body to expose at least a part of the first guide at an opened state of the cover. The second guide is in the cover and faces the first guide to form a sheet conveyance path at a closed state of the cover. The detector detects a sheet passing through the sheet conveyance path. The detector includes a photosensor in the body and a feeler in the cover. The feeler is movable relative to the second guide. The feeler includes a detected portion and a contacted portion. The detected portion is movable in conjunction with the sheet passing through the sheet conveyance path. The contacted portion is attached to the detected portion. The photosensor has a detection region in which the detected portion entering the detection region is detected by the photosensor. The positioner is in the body to contact the contacted portion to position the detected portion in the detection region when the sheet is not in the sheet conveyance path.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] A more complete appreciation of embodiments of the present disclosure and many of the attendant advantages and features thereof can be readily obtained and understood from the following detailed description with reference to the accompanying drawings, wherein:

[0006] FIG. 1 is a diagram illustrating an overall configuration of an image forming apparatus;

[0007] FIGS. 2A and 2B are schematic diagrams of the image forming apparatus of FIG. 1 to illustrate an operation to open a cover;

[0008] FIG. 3A is a diagram illustrating a detector when a sheet does not pass through a sheet

conveyance path;

[0009] FIG. 3B is a diagram illustrating the detector of FIG. 3A when a sheet passes through the sheet conveyance path of FIG. 3A;

[0010] FIG. 4 is a diagram illustrating the detector of FIGS. 3A and 3B when the cover of FIGS. 3A and 3B is opened;

[0011] FIG. 5A is a diagram illustrating a rotatable angle of a feeler when a cover is closed;

[0012] FIG. 5B is a diagram illustrating a rotatable angle of a feeler when a cover is opened;

[0013] FIGS. 6A and 6B are diagrams illustrating a body side guide and a cover side guide that extend in a width direction;

[0014] FIG. 7A is a diagram illustrating a detector in a sheet conveying device according to a comparative example when a cover is not deformed;

[0015] FIG. 7B is a diagram illustrating the detector in the sheet conveying device according to the comparative example when the cover is deformed in FIG. 7A;

[0016] FIG. 8 is a diagram illustrating the detector of FIG. 7A when the cover of FIG. 7A is opened; and

[0017] FIG. 9 is a diagram illustrating an overall configuration of an image forming system as a modification.

[0018] The accompanying drawings are intended to depict embodiments of the present disclosure and should not be interpreted to limit the scope thereof. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. Also, identical or similar reference numerals designate identical or similar components throughout the several views.

DETAILED DESCRIPTION

[0019] In describing embodiments illustrated in the drawings, specific terminology is employed for the sake of clarity. However, the disclosure of this specification is not intended to be limited to the specific terminology so selected and it is to be understood that each specific element includes all technical equivalents that have a similar function, operate in a similar manner, and achieve a similar result.

[0020] Referring now to the drawings, embodiments of the present disclosure are described below. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

[0021] Embodiments of the present disclosure are described below in detail with reference to the drawings. Like reference signs are assigned to identical or equivalent components and a description of those components may be simplified or omitted.

[0022] Initially with reference to FIG. 1, a configuration and an operation of an image forming apparatus 1 are described below.

[0023] As illustrated in FIG. 1, the image forming apparatus 1 according to the present embodiment is a tandem-type color printer. The image forming apparatus 1 includes a bottle housing 101 in an upper portion of the image forming apparatus 1. The bottle housing 101 accommodates four toner bottles 102Y, 102M, 102C, and 102K containing fresh yellow, magenta, cyan, and black toners, respectively. The four toner bottles 102Y, 102M, 102C, and 102K are detachably attached to the bottle housing 101 for replacement.

[0024] Under the bottle housing 101, an intermediate transfer unit 85 is disposed. Facing an intermediate transfer belt 78 (an intermediate transferor) of the intermediate transfer unit 85, image forming devices 4Y, 4M, 4C, and 4K are arranged side by side to form toner images of yellow, magenta, cyan, and black, respectively.

[0025] The image forming apparatus 1 includes a sheet feeder 12 (a sheet tray) in a lower portion of the body of the image forming apparatus 1. The sheet feeder 12 contains a stack of multiple sheets P such as sheets of paper stacked on one on another.

[0026] In addition, a sheet conveying device 49 is disposed on the right side of the body of the image forming apparatus 1. The sheet conveying device 49 includes a part of the body of the image

forming apparatus **1**. In the following embodiment, the body of the image forming apparatus **1** is described as one example of a body of the sheet conveying device **49**. The sheet conveying device **49** conveys and ejects the sheet P toward a stack portion **100** or reverses (switches back) the sheet P and conveys the sheet P through a duplex printing conveyance path K5. In the sheet conveying device **49**, a part of the duplex printing sheet conveyor and a secondary transfer roller **89** described below are disposed on a cover **50** that can be opened and closed.

[0027] The image forming devices **4Y**, **4M**, **4C**, and **4K** include photoconductor drums **5Y**, **5M**, **5C**, and **5K**, respectively. Each of the photoconductor drums **5Y**, **5M**, **5C**, and **5K** is surrounded by a charger **75**, a developing device **76**, a cleaner **77**, and a discharger. Image forming processes including a charging process, an exposure process, a developing process, a primary transfer process, and a cleaning process are performed on each of the photoconductor drums **5Y**, **5M**, **5C**, and **5K**, forming yellow, magenta, cyan, and black toner images on the photoconductor drums **5Y**, **5M**, **5C**, and **5K**, respectively.

[0028] A main motor drives to rotate the photoconductor drums **5Y**, **5M**, **5C**, and **5K** clockwise in FIG. **1**. The chargers **75** uniformly charge the surfaces of the photoconductor drums **5Y**, **5M**, **5C**, and **5K**, respectively, which is referred to as the charging process.

[0029] After the charging process, the charged outer circumferential surface of each of the photoconductor drums **5Y**, **5M**, **5C**, and **5K** reaches an irradiation position at which an exposure device **3** (in other words, a writing device) irradiates and scans the photoconductor drums **5Y**, **5M**, **5C**, and **5K** with laser beams, irradiating and scanning the photoconductor drums **5Y**, **5M**, **5C**, and **5K** with the laser beams forms electrostatic latent images according to yellow, magenta, cyan, and black image data in the exposure process.

[0030] After the exposure process, the irradiated and scanned outer circumferential surface of each of the photoconductor drums **5Y**, **5M**, **5C**, and **5K** reaches a developing position at which the developing device **76** is disposed opposite each of the photoconductor drums **5Y**, **5M**, **5C**, and **5K**, and the developing device **76** develops the electrostatic latent image formed on the respective photoconductor drums **5Y**, **5M**, **5C**, and **5K**, thus forming yellow, magenta, cyan, and black toner images on the photoconductor drums **5Y**, **5M**, **5C**, and **5K** in the developing process.

[0031] After the developing process, the yellow, magenta, cyan, and black toner images formed on the photoconductor drums **5Y**, **5M**, **5C**, and **5K** reach primary transfer nips formed between the photoconductor drums **5Y**, **5M**, **5C**, and **5K** and the intermediate transfer belt **78** by four primary transfer bias rollers **79Y**, **79M**, **79C**, and **79K** pressed against the four photoconductor drums **5Y**, **5M**, **5C**, and **5K** via the intermediate transfer belt **78**, respectively, and the yellow, magenta, cyan, and black toner images are primarily transferred onto the intermediate transfer belt **78** as the intermediate transferor in a primary transfer process. After the primary transfer process, residual toner failed to be transferred onto the intermediate transfer belt **78** remains on the photoconductor drums **5Y**, **5M**, **5C**, and **5K** slightly.

[0032] After the primary transfer process, the residual toner on each of the photoconductor drums **5Y**, **5M**, **5C**, and **5K** reaches a cleaning position at which the cleaner **77** is disposed opposite each of the photoconductor drums **5Y**, **5M**, **5C**, and **5K**, and a cleaning blade of the cleaner **77** mechanically collects the residual toner from each of the photoconductor drums **5Y**, **5M**, **5C**, and **5K** in the cleaning process.

[0033] Finally, the cleaned outer circumferential surface of each of the photoconductor drums **5Y**, **5M**, **5C**, and **5K** reaches a discharging position at which the discharger is disposed opposite each of the photoconductor drums **5Y**, **5M**, **5C**, and **5K**, and the discharger eliminates residual potential from each of the photoconductor drums **5Y**, **5M**, **5C**, and **5K**.

[0034] Thus, a series of image forming processes performed on the photoconductor drums **5Y**, **5M**, **5C**, and **5K** is finished.

[0035] The yellow, magenta, cyan, and black toner images formed on the photoconductor drums **5Y**, **5M**, **5C**, and **5K** in the developing process are primarily transferred onto an outer

circumferential surface of the intermediate transfer belt **78** such that the yellow, magenta, cyan, and black toner images are superimposed on the intermediate transfer belt **78**. Thus, a color toner image is formed on the surface of the intermediate transfer belt **78**.

[0036] The intermediate transfer unit **85** includes the intermediate transfer belt **78** as the intermediate transferor, the four primary transfer bias rollers **79Y**, **79M**, **79C**, and **79K**, a secondary transfer backup roller **82**, a cleaning backup roller **83**, a tension roller **84**, and an intermediate transfer belt cleaner **80**. The intermediate transfer belt **78** is stretched taut across and supported by the three rollers, that is, the secondary transfer backup roller **82**, the cleaning backup roller **83**, and the tension roller **84**. One of the three rollers, that is, the secondary transfer backup roller **82** is coupled to the main motor to drive and rotate the intermediate transfer belt **78** in a rotation direction indicated by an arrow in FIG. **1**.

[0037] The four primary transfer bias rollers **79Y**, **79M**, **79C**, and **79K** sandwich the intermediate transfer belt **78** together with the four photoconductor drums **5Y**, **5M**, **5C**, and **5K**, respectively, thus forming the four primary transfer nips between the intermediate transfer belt **78** and the photoconductor drums **5Y**, **5M**, **5C**, and **5K**. A primary transfer bias opposite in polarity to the toner is applied to the primary transfer bias rollers **79Y**, **79M**, **79C**, and **79K**.

[0038] The intermediate transfer belt **78** is moved in the direction indicated by the arrow in FIG. **1** and sequentially passes through the primary transfer nips formed by the primary transfer bias rollers **79Y**, **79M**, **79C**, and **79K**. Thus, the toner images formed on the respective photoconductor drums **5Y**, **5M**, **5C**, and **5K** are primarily transferred onto the intermediate transfer belt **78** in a manner of being superimposed one atop another to form a composite color toner image on the intermediate transfer belt **78** in the primary transfer process.

[0039] Subsequently, the intermediate transfer belt **78** bearing the composite color toner image reaches a position opposite a secondary transfer roller **89** as a transfer device. At the position, the secondary transfer backup roller **82** and the secondary transfer roller **89** (the transfer device) press against each other via the intermediate transfer belt **78**, and the contact portion therebetween is referred to as a secondary transfer nip below. Specifically, the secondary transfer roller **89** is in contact with the secondary transfer backup roller **82** via the intermediate transfer belt **78** at a predetermined pressure (a contact pressure).

[0040] At the secondary transfer nip, the composite color toner image (or four-color toner image including yellow, magenta, cyan, and black colors) is secondarily transferred from the intermediate transfer belt **78** onto a sheet **P** conveyed to the position of the secondary transfer nip, in a secondary transfer process. At this time, residual toner that is not transferred onto the sheet **P** remains on the surface of the intermediate transfer belt **78**.

[0041] The surface of the intermediate transfer belt **78** then reaches a position opposite the intermediate transfer belt cleaner **80**. At the position, the intermediate transfer belt cleaner **80** collects the residual toner from the intermediate transfer belt **78**.

[0042] Thus, a series of transfer processes performed on the intermediate transfer belt **78** is completed.

[0043] The sheet **P** is conveyed from the sheet feeder **12** disposed in the lower portion of the body of the image forming apparatus **1** to the secondary transfer nip via a first sheet conveyance path **K0** and a second sheet conveyance path **K1**.

[0044] Specifically, the sheet feeder **12** contains a stack of multiple sheets **P** such as sheets of paper stacked on top of one another. An uppermost sheet **P** of the multiple sheets **P** is nipped between a feed roller **31** and a friction pad **32**. Rotating the feed roller **31** counterclockwise in FIG. **1** feeds the uppermost sheet **P** toward a portion between a registration roller pair **33** as a timing roller pair. Guide plates forming the first sheet conveyance path **K0** and the second sheet conveyance path **K1** guide the uppermost sheet **P** to the portion between the registration roller pair **33**.

[0045] The sheet **P** conveyed to the registration roller pair **33** serving as the timing roller pair temporarily stops at a position of a roller nip (a nip) of the registration roller pair **33** that stops

rotating. Subsequently, the registration roller pair **33** rotates to convey the sheet P to the secondary transfer nip in an image forming section, timed to coincide with the arrival of the color toner image on the intermediate transfer belt **78**. Thus, the desired color toner image is transferred onto the sheet P.

[0046] After the secondary transfer roller **89** transfers the color toner image onto the sheet P at the secondary transfer nip in the image forming section, the sheet P is conveyed to a fixing device **20** via the second sheet conveyance path **K1**. In the fixing device **20**, a fixing belt **21** and a pressure roller **22** apply heat and pressure to the sheet P at a fixing nip formed by the fixing belt **21** and the pressure roller **22** pressing each other to fix the color toner image on the sheet P, which is referred to as a fixing process.

[0047] After the fixing process, the sheet P passes through the second sheet conveyance path **K1** and is guided to a sheet ejection conveyance path **K2** by a bifurcating claw **45** as a switching member, and a first roller **41** and a second roller **40** that function as an ejection roller pair eject the sheet P to the outside of the image forming apparatus **1**. The sheets P ejected by the first roller **41** and the second roller **40** are sequentially stacked as output images on the stack portion **100**.

[0048] Thus, a series of image forming processes performed by the image forming apparatus **1** is completed.

[0049] The above-described operations of the image forming apparatus **1** and movement of the sheet P from the sheet feeder **12** to the stack portion **100** are performed when a single-sided print mode is selected. In the single-sided print mode, the image is formed only on the front face of the sheet P.

[0050] In the single-sided print mode (or at the end of a double-sided print mode when the sheet P is ejected to the stack portion **100** after the images are printed on both sides of the sheet P), the bifurcating claw **45** as the switching member is at a position (the position illustrated in FIG. **1**) after the bifurcating claw **45** rotates counterclockwise about the rotation shaft so as to open the sheet ejection conveyance path **K2** and close a relay sheet conveyance path **K3**.

[0051] When the double-sided print mode (a mode in which images are formed on the front face and the back face of the sheet P, respectively) is selected, the image forming apparatus **1** operates as follows, and the sheet P moves as follows.

[0052] The processes until the sheet P fed from the sheet feeder **12** reaches the fixing device **20** via the first and second sheet conveyance paths **K0** and **K1** and the secondary transfer nip are the same as those in the single-sided print mode. After the fixing process (after the image is formed on the front face of the sheet P), the bifurcating claw **45** as the switching member rotates clockwise about the rotation shaft and stops at a position to close the sheet ejection conveyance path **K2** and open the relay sheet conveyance path **K3**. The bifurcating claw **45** as the switching member guides the sheet P to a switchback sheet conveyance path **K4** via the relay sheet conveyance path **K3**.

[0053] In the switchback sheet conveyance path **K4**, a reverse roller **42** temporarily stops rotating when the rear end (the rear end in a conveyance direction) of the sheet P reaches the nip of the reverse roller **42** and the second roller **40** after the rear end of the sheet P passes through the branch point between the relay sheet conveyance path **K3** and the duplex printing conveyance path **K5**. At this time, the rear end of the sheet P having a fixed image on the front face is held by the reverse roller **42**, and the front end of the sheet P is above the stack portion **100** and exposed to the outside of the image forming apparatus **1**.

[0054] Subsequently, rotating the reverse roller **42** in reverse reverses the conveyance direction of the sheet P and conveys the sheet P toward the double-sided sheet conveyance path **K5**. At this time, the bifurcating claw **45** as the switching member rotates counterclockwise about the rotation shaft and stops at the position to close the relay sheet conveyance path **K3** and open the sheet ejection conveyance path **K2** and the double-sided sheet conveyance path **K5** as illustrated in FIG. **1**.

[0055] Referring to FIG. **1**, the sheet P guided to the double-sided sheet conveyance path **K5** is

conveyed by a plurality of conveying rollers **47** and **48** disposed in the double-sided sheet conveyance path **K5** and guided to the secondary transfer nip. At the secondary transfer nip, an image is secondarily transferred to the back face of the sheet **P** similar to the secondary transfer process for the front face of the sheet **P**. After the secondary transfer process, the sheet **P** is conveyed to the fixing device **20** to perform the fixing process for the back face of the sheet **P**. [0056] After the fixing process (that is, after the images are printed on both sides of the sheet **P**), the sheet **P** is guided to the nip of the first roller **41** and the second roller **40** via the sheet ejection conveyance path **K2**, and the first roller **41** and the second roller **40** ejects the sheet **P** to the outside of the image forming apparatus **1** as described above. As a result, the sheets **P** are sequentially stacked on the stack portion **100**.

[0057] The fixing device **20** includes the fixing belt **21**, the pressure roller **22**, and a planar heater **23**.

[0058] The fixing belt **21** is an endless belt including a base layer and a release layer as a surface layer layered on the base layer. The base layer is made of metal such as nickel or stainless steel or resin such as polyimide. The release layer is made of tetrafluoroethylene-perfluoroalkylvinylether copolymer (PFA), polytetrafluoroethylene (PTFE), polyimide, polyetherimide, or polyethersulfone (PES). The fixing belt **21** according to the present embodiment does not include an elastic layer made of rubber between the base layer and the release layer as the surface layer to reduce a manufacturing cost.

[0059] Inside a loop of the fixing belt **21**, the planar heater **23** as a heat source is disposed. The planar heater **23** contacts the inner circumferential surface of the fixing belt **21**. The planar heater **23** is pressed against the pressure roller **22** via the fixing belt **21** to form the fixing nip through which the sheet **P** is conveyed. The planar heater **23** disposed inside the loop of the fixing belt **21** directly heats the fixing belt **21**. The outer circumferential surface of the fixing belt **21** heated by the planar heater **23** heats the toner image on the sheet **P**.

[0060] The pressure roller **22** includes a cored bar and an elastic layer made of rubber and formed on the cored bar.

[0061] The following describes a characteristic configuration and operation of the sheet conveying device **49** installed in the image forming apparatus **1** in detail with reference to FIGS. **2A** to **6B**.

[0062] The sheet conveying device **49** conveys the sheet **P**.

[0063] As illustrated in FIGS. **2A** and **2B**, the sheet conveying device **49** includes the cover **50** on the body of the image forming apparatus **1**. The cover **50** rotates about a support shaft **50a** attached to the body of the image forming apparatus **1** to open and close the inside of the body of the image forming apparatus **1**.

[0064] Specifically, the cover **50** is closed as illustrated in FIGS. **1**, **2A**, and **3A**, which is referred to as a closed state of the cover **50**, and rotated clockwise (in the direction indicated by the arrow in FIG. **2B**) about the support shaft **50a** to open the cover **50** as illustrated in FIGS. **2B** and **4**, which is referred to as an opened state of the cover **50**, and opening the cover **50** opens and exposes a part of the inside of the body of the image forming apparatus **1** such as the first sheet conveyance path **K0**, the second sheet conveyance path **K1**, the fixing device **20**, and the intermediate transfer unit **85**. As a result, the above-described configuration enables removing the sheet jammed in the first sheet conveyance path **K0** or the second sheet conveyance path **K1**, detaching the fixing device **20** or the intermediate transfer unit **85**, performing maintenance of the fixing device **20** or the intermediate transfer unit **85**, and attaching the fixing device **20** or the intermediate transfer unit **85**. Rotating the cover **50** opened as illustrated in FIGS. **2B** and **4** counterclockwise in FIGS. **2B** and **4** about the support shaft **50a** closes the cover **50** as illustrated in FIGS. **1**, **2A**, and **3A**. As a result, the image forming apparatus can perform the printing operation described above with reference to FIG. **1**.

[0065] The closed state of the cover **50** means that the body of the image forming apparatus **1** is closed by the cover **50**. On the other hand, the opened state of the cover **50** means that the body of the image forming apparatus **1** is opened by the cover **50**.

[0066] When a user sets the sheets P to the sheet feeder **12** or takes out the sheet P printed from the stack portion **100**, the user stands at the front side as an operation side of the image forming apparatus **1**. The cover **50** is disposed on the right side or the left side of the body of the image forming apparatus **1** and functions as a side cover.

[0067] In the sheet conveying device **49** or the image forming apparatus **1** as described above and will be described later with reference to FIGS. **1** to **4**, the second sheet conveyance path K**1** as a sheet conveyance path is arranged downstream from the fixing device **20** in the conveyance direction. Two guides (a body side guide **54** and a cover side guide **55**) face each other with a space through which the sheet P can pass to form the second sheet conveyance path K**1**.

[0068] The body side guide **54** as a first guide is in the body of the image forming apparatus **1**. The cover side guide **55** as a second guide is in the cover **50** and faces the body side guide **54** to form the second sheet conveyance path K**1** as the sheet conveyance path.

[0069] In the present embodiment, each of the body side guide **54** and the cover side guide **55** may have a plate shape curved to guide the sheet P in the sheet conveyance direction and multiple ribs extending in the sheet conveyance direction on the surface of the plate shape.

[0070] The cover **50** including the cover side guide **55** as the second guide rotates together with the cover side guide **55** about the support shaft **50a** with respect to the body of the image forming apparatus **1** to expose a part of or an entire of the body side guide **54** as the first guide.

[0071] In other words, rotating the cover **50** together with the cover side guide **55** about the support shaft **50a** and opening the cover **50** including the cover side guide **55** as illustrated in FIGS. **2B** and **4** separates the cover side guide **55** from the body side guide **54** fixed on the body of the image forming apparatus **1**. As a result, the user can easily remove the sheet P jammed (stopped) in the vicinity of the second sheet conveyance path K**1**.

[0072] In addition, opening the cover **50** as illustrated in FIGS. **2B** and **4** exposes the fixing device **20** installed in the image forming apparatus **1**. This enables jam processing for the sheet P jammed at the fixing nip of the fixing device **20** and maintenance of the fixing device **20**.

[0073] With reference to FIG. **3** and FIG. **1**, a detector **90** is disposed adjacent to the second sheet conveyance path K**1** formed by the sheet conveying device **49** (the image forming apparatus **1**). The detector **90** optically detects whether the sheet P passing through the second sheet conveyance path K**1** is in the second sheet conveyance path K**1** as the sheet conveyance path.

[0074] With reference to FIGS. **3A** to **5B**, the detector **90** includes a transmission-type photosensor **91** as a photosensor, a feeler **92** (a rotator), and a torsion spring **93** as a biasing member.

[0075] The transmission-type photosensor **91** as the photosensor is not in the cover **50** but is in the body of the image forming apparatus **1**. The transmission-type photosensor **91** is always fixed to the body of the image forming apparatus **1** regardless of the opening and closing of the cover **50**. The transmission-type photosensor **91** includes a light emitting element and a light receiving element that face each other. A gap between the light emitting element and the light receiving element is designed so that the feeler **92** can come in and out of the gap.

[0076] The feeler **92** is disposed to be rotatable about a rotation shaft **92a** with respect to the cover side guide **55**. In other words, the feeler **92** is not in the body of the image forming apparatus **1** but in the cover **50**. Therefore, the feeler **92** rotates about the support shaft **50a** together with the cover **50** in accordance with the opening and closing of the cover **50**.

[0077] Since a power supply to supply power to the transmission-type photosensor **91** is in the body of the image forming apparatus **1**, the transmission-type photosensor **91** is in the body of the image forming apparatus **1** in order to continue power supply from the power supply to the transmission-type photosensor **91** regardless of the opening and closing of the cover **50**. In order to prevent the feeler **92** from contacting and rubbing the image fixed on the sheet P, the feeler **92** is disposed on the cover side guide **55** in the cover **50**. Further, to detect the opening and closing of the cover **50**, the detector **90** is configured as described above.

[0078] The transmission-type photosensor **91** has a detection region **91a** that is defined as a region

in which the transmission-type photosensor **91** can detect whether the feeler **92** is in the region. As illustrated in FIGS. **3A** and **3B**, the feeler **92** in the closed cover **50** is rotated in conjunction with the passage of the sheet **P**. As a result, the feeler **92** comes in and out of the detection region **91a**. [0079] In detail, the feeler **92** has a detected portion **92c**. The biasing force of the torsion spring **93** biases the feeler **92** so as to rotate the feeler **92** counterclockwise in FIG. **3A**. As illustrated in FIG. **3A**, the detected portion **92c** is in the detection region **91a** of the transmission-type photosensor **91** when the sheet **P** is not in the second sheet conveyance path **K1** between the body side guide **54** and the cover side guide **55**. As a result, the detected portion **92c** blocks light emitted from the light emitting element to the light receiving element in the transmission-type photosensor **91**. Blocking the light causes the light receiving element of the transmission-type photosensor **91** to transmit an ON signal (or an OFF signal) to a controller, and the controller determines that the sheet **P** is not in the second sheet conveyance path **K1**.

[0080] In contrast, as illustrated in FIG. **3B**, the sheet **P** in the second sheet conveyance path **K1** between the body side guide **54** and the cover side guide **55** pushes the feeler **92** and rotates the feeler **92** clockwise in FIG. **3B** about the rotation shaft **92a** against the biasing force of the torsion spring **93**. As a result, the detected portion **92c** moves to the outside of the detection region **91a** of the transmission-type photosensor **91**, and the light emitted from the light emitting element is emitted to the light receiving element. The light receiving element receiving the light in the transmission-type photosensor **91** transmits the OFF signal (or the ON signal) to the controller, and the controller determines that the sheet **P** is in the second sheet conveyance path **K1**. In other words, the detector **90** includes the feeler **92** movable relative to the cover side guide **55**, and the feeler **92** includes the detected portion **92c** movable in conjunction with the sheet passing through the second sheet conveyance path **K1**. The transmission-type photosensor **91** has the detection region **91a** in which the detected portion **92c** entering the detection region **91a** is detected by the transmission-type photosensor **91**. As a result, the detector **90** detects the sheet passing through the second sheet conveyance path **K1**.

[0081] The controller uses the above-described results detected by the detector **90** regarding whether the sheet **P** is in the sheet conveyance path **K1** to check the timing of passage of the sheet **P** in the second sheet conveyance path **K1**, determine whether the sheet **P** passes the second sheet conveyance path **K1**, and determine whether the sheet **P** is jammed in the second sheet conveyance path **K1**.

[0082] The sheet conveying device **49** of the image forming apparatus **1** includes a positioner **59**. As illustrated in FIGS. **3A** and **5A**, when the sheet **P** is not in the second sheet conveyance path **K1** as the sheet conveyance path formed by the closed cover **50**, the positioner **59** contacts a contacted portion **92d** of the feeler **92** and positions the feeler **92** with respect to the transmission-type photosensor **91**.

[0083] The positioner **59** has a pin shape and is in the body of the image forming apparatus **1**, not in the cover **50**. Regardless of the opening and closing of the cover **50**, the positioner **59** is fixed in the body of the image forming apparatus **1**. Specifically, the positioner **59** in the body of the image forming apparatus **1** contacts the contacted portion **92d** of the feeler **92** to position the detected portion **92c** of the feeler **92** in the detection region **91a** of the transmission-type photosensor **91** when the sheet is not in the second sheet conveyance path **K1**. The position at which the positioner **59** is fixed in the body is designed so as not to block the sheet **P** conveyed in the sheet conveyance path.

[0084] Specifically, as illustrated in FIG. **5A**, the feeler **92** has a substantial V-shape or a substantial boomerang-shape like a member bent about the rotation shaft **92a**. The feeler **92** includes the rotation shaft **92a**, a lever **92b**, the detected portion **92c**, the contacted portion **92d** as a first contacted portion, and a second contacted portion **92f** as another contacted portion.

[0085] The lever **92b** projects from the rotation shaft **92a** to contact the sheet **P** passing through the second sheet conveyance path **K1** as the sheet conveyance path. As described above with reference

to FIG. 3B, the sheet P pushes the lever **92b**, and thus the feeler **92** rotates about the rotation shaft **92a**. The cover side guide **55** has a sheet guide face having an opening. The lever **92b** of the feeler **92** protrudes from the opening toward the body side guide **54** (in other words, the second sheet conveyance path **K1**).

[0086] The detected portion **92c** is disposed at a position away from the lever **92b** via the rotation shaft **92a**. The rotation shaft **92a** is interposed between the detected portion **92c** and the lever **92b**. In other words, the detected portion **92c** is separated from the lever **92b** and the rotation shaft **92a** by the second contacted portion **92f**. The detected portion **92c** comes in and out of the detection region **91a** of the transmission-type photosensor **91**. As described above with reference to FIGS. 3A and 3B, rotating the feeler **92** forward and reverse about the rotation shaft **92a** causes the detected portion **92c** of the feeler **92** to come in and out of the detection region **91a**. As a result, the light emitted from the light emitting element is blocked by the detected portion **92c** or emitted to the light receiving element.

[0087] The contacted portion **92d** as the first contacted portion is positioned to be farther from the rotation shaft **92a** than the detected portion **92c**. The contacted portion **92d** contacts the positioner **59** and receives the biasing force of the torsion spring **93** to prevent the feeler **92** from rotating too much about the rotation shaft **92a** counterclockwise in FIG. 5A.

[0088] Integrally forming the contacted portion **92d** and the detected portion **92c** enhances the positional accuracy of the detected portion **92d** with respect to the detection region **91a** when the cover **50** is closed (in other words, when the contacted portion **92d** contacts the positioner **59**).

[0089] The second contacted portion **92f** functions as a portion that relays the rotation shaft **92a** and the detected portion **92c** and is configured to contact a second positioner **55a** as another positioner of the cover side guide **55** to position the feeler **92** with respect to the cover side guide **55** at the opened state of the cover **50**.

[0090] On the other hand, the cover side guide **55** includes the second positioner **55a** and the torsion spring **93**.

[0091] The second positioner **55a** has a pin shape and is formed integrally with the cover side guide **55**. After the cover **50** is opened, the second positioner **55a** contacts the second contacted portion **92f** of the feeler **92** to determine the position of the feeler **92** with respect to the cover side guide **55** as illustrated in FIG. 5B. The second contacted portion **92f** contacts the second positioner **55a** and receives the biasing force of the torsion spring **93** to prevent the feeler **92** from rotating too much about the rotation shaft **92a** counterclockwise in FIG. 5B.

[0092] The torsion spring **93** functions as a biasing member that biases the feeler **92** so that the feeler **92** rotates and contacts the positioner **59** (or the second positioner **55a**). The torsion spring **93** has a coil portion inserted into the rotation shaft **92a**, one arm coupled to the cover side guide **55**, and the other arm coupled to a portion of the feeler **92** that is the portion away from the rotation shaft **92a** such as the detected portion **92c**. The biasing force of the torsion spring **93** is set to a magnitude such that the sheet P pushing the lever **92b** can rotate the feeler **92** clockwise in FIG. 5 about the rotation shaft **92a**, and to a magnitude such that the feeler **92** rotates counterclockwise in FIG. 5 about the rotation shaft **92a** after the sheet P does not push the lever **92b**.

[0093] After the cover **50** is closed, the feeler **92** can rotate at an angle α as illustrated in FIG. 5A. After the cover **50** is opened, the feeler **92** can rotate at an angle β as illustrated in FIG. 5B. The angle α is designed to be smaller than the angle β ($\alpha < \beta$). In other words, a rotatable angle α of the feeler **92** at the closed state of the cover **50** is designed to be smaller than a rotatable angle β of the feeler **92** at the opened state of the cover **50**.

[0094] The above-described configuration causes the feeler **92** to reliably contact the first positioner **59** while the opened cover **50** (in which the feeler **92** contacts the second positioner **55a**) is closed, which reliably determines the position of the feeler **92** with respect to the transmission-type photosensor **91**. As a result, the above-described configuration prevents the detector from causing erroneous detection due to poor positional accuracy of the feeler **92** with respect to the

transmission-type photosensor **91**.

[0095] As described above, the positioner **59** disposed in the body of the image forming apparatus **1** determines the position of the feeler **92** of the detector **90** disposed in the cover **50** with respect to the transmission-type photosensor **91** of the detector **90** disposed in the body of the image forming apparatus **1**.

[0096] The above-described configuration easily determines the position of the feeler **92** in the closed cover **50** with respect to the transmission-type photosensor **91** even if the cover **50** has deformation such as warping.

[0097] Specifically, a comparative example is described below with reference to FIGS. 7A, 7B, and **8**. In the comparative example, a positioner **55x** disposed in the cover side guide **55** contacts the feeler **192** in the closed cover **50** to determine the position of the feeler **192** with respect to the transmission-type photosensor **91** (see FIGS. 7A and **8**). As illustrated in FIG. 7B, the deformation of the cover **50** such as the warping is likely to cause the deviation of the detected portion **92c** of the feeler **192** in the closed cover **50** from the detection region **91a** of the transmission-type photosensor **91**. In such a case, the position of the feeler **92** with respect to the transmission-type photosensor **91** is not correctly fixed, and the detector **90** cannot sufficiently detect the passage of the sheet P or the occurrence of the sheet jam in the second sheet conveyance path **K1**.

[0098] In contrast, since the positioner **59** in the present embodiment is disposed in the body of the image forming apparatus **1** and determines the position of the feeler **92** in the closed cover **50** with respect to the transmission-type photosensor **91**, the positional accuracy of the feeler **92** in the closed cover **50** with respect to the transmission-type photosensor **91** can be enhanced even if the cover **50** is deformed.

[0099] In particular, the fixing device **20** includes the fixing belt **21** including no elastic layer as described above. Such a fixing belt has low strength. If the detector **90** has a low jam detection accuracy, the sheet jam is likely to occur, and the jammed sheet is wound around the fixing belt **21**. In such a case, the jammed sheet P presses the planar heater **23** via the fixing belt **21** and may break the planar heater **23**.

[0100] In contrast, the detector **90** according to the present embodiment has a high positional accuracy of the feeler **92** with respect to the transmission-type photosensor **91**, and thus the jam of the sheet P is unlikely to be erroneously detected. As a result, the above-described problem is unlikely to occur.

[0101] Preferably, the transmission-type photosensor **91** and the positioner **59** are disposed at the body side guide **54**.

[0102] The above-described configuration can enhance the positional accuracy of the detector **90** with respect to the body side guide **54**.

[0103] The transmission-type photosensor **91** and the positioner **59** may be integrally formed as a single component. The above-described configuration can further enhance the positional accuracy of the detector **90** because a dimensional error or an assembly error is less likely to occur in the positional accuracy of the positioner **59** with respect to the transmission-type photosensor **91**.

[0104] As illustrated in FIG. 6A, the cover side guide **55** in the image forming apparatus **1** includes an engaging portion **97**, and the body side guide **54** in the image forming apparatus **1** includes an engaged portion **96**. Closing the cover **50** causes the engaging portion **97** to engage with the engaged portion **96**, which determines the position of the cover side guide **55** with respect to the body side guide **54** in a width direction. The width direction of the cover side guide **55** is the same as the direction perpendicular to each of the surfaces of the paper on which FIGS. **1** to **4** are drawn, the lateral direction in FIG. 6A, and the direction orthogonal to the sheet conveyance direction in which the sheet P passes through the second conveyance path **K1** and in parallel to the surface of the sheet P.

[0105] FIG. 6B is a schematic diagram illustrating a comparative example not including the above-described positioning mechanism in the width direction. If the deformation occurs in the cover **50**,

the position of the cover side guide **55** in the width direction is shifted with respect to the body side guide **54**. Closing the cover **50** deformed as described above causes the feeler **92** disposed on the cover side guide **55** to ride on the transmission-type photosensor **91** disposed on the body side guide **54** in the body of the image forming apparatus **1** as illustrated in the circle drawn by a dashed line in FIG. **6B**. As a result, the detector **90** does not operate normally.

[0106] In contrast, in the present embodiment, the cover side guide **55** includes the engaging portion **97**, and the body side guide **54** includes the engaged portion **96**, which serve as the positioning mechanism in the width direction as illustrated in FIG. **6A**. Even if the cover **50** having the deformation that shifts the position of the cover side guide **55** in the width direction with respect to the body side guide **54** is closed, the engaging portion **97** (a striker) engages with the engaged portion **96** having an opening expanding in a conical shape in the width direction to correct the shift of the cover side guide **55**. As a result, the above-described configuration in the present embodiment overcomes the disadvantage that the feeler **92** disposed on the cover side guide **55** rides on the transmission-type photosensor **91** disposed on the body side guide **54** in the body of the image forming apparatus **1**. The position of the feeler **92** in the width direction with respect to the transmission-type photosensor **91** is accurately determined, and the detector **90** operates under normal operating conditions.

[0107] With reference to FIGS. **3A**, **3B**, and **4**, the cover **50** includes a reinforcement **94** to reinforce the strength of the cover **50**. The reinforcement **94** is in a region closer to the support shaft **50a** than the position of the feeler **92**. In other words, the reinforcement **94** in the cover **50** is closer to the support shaft **50a** than to the feeler **92**. The reinforcement **94** is made of, for example, a metal plate to reinforce the mechanical strength of the cover **50** made of resin and prevent the cover **50** from being deformed. The more the reinforcement **94** reinforces the entire area of the cover **50**, the more the purpose of use is achieved, but the weight of the cover **50** is increased.

[0108] As described above, in the present embodiment, the positioner **59** disposed in the body of the image forming apparatus **1** determines the position of the feeler **92** of the detector **90** disposed in the cover **50** with respect to the transmission-type photosensor **91** of the detector **90** disposed in the body of the image forming apparatus **1**. The above-described configuration easily determines the position of the feeler **92** in the closed cover **50** with respect to the transmission-type photosensor **91** even if the cover **50** has the deformation.

[0109] Accordingly, the reinforcement **94** in the present embodiment is in a region between the support shaft **50a** in the cover **50** and the feeler **92** (the cover side guide **55**) as an optimum region. In the optimum region, the positional accuracy of the feeler **92** with respect to the transmission-type photosensor **91**, the increase in the strength of the cover **50**, and weight reduction of the cover **50** can be achieved in a well-balanced manner.

[0110] A description is given of a modification.

[0111] As illustrated in FIG. **9**, an image forming system **200** as the modification includes an inkjet printer **99** as the image forming apparatus, a coating apparatus **150** as a sheet processing apparatus attached to the inkjet printer **99** to perform pre-processing, a dryer **185** attached to the inkjet printer **99** to perform post-processing.

[0112] FIG. **9** is a diagram illustrating the inkjet printer **99** as the image forming apparatus; the coating apparatus **150** to coat coating liquid as the pre-processing on the sheet **P** conveyed to the image forming apparatus **1**; a sheet feeder **180** to feed the sheet **P** such as a paper sheet; the dryer **185** to dry ink on the sheet **P** as the post-processing on the sheet **P** after image formation; and a sheet ejection device **190** to stack the sheet **P** ejected from the dryer **185**.

[0113] As illustrated in FIG. **9**, the image forming system **200** according to the present modification includes the sheet feeder **180**, the coating apparatus **150**, the inkjet printer **99** as the image forming apparatus, the dryer **185**, and the sheet ejection device **190** that are connected and arranged in this order from upstream to downstream in the sheet conveyance direction.

[0114] With reference to FIG. **9**, the operation of the image forming system **200** is briefly

described.

[0115] In response to inputting a print command together with image data from a personal computer to a controller of the image forming system **200**, a feed roller **182** feeds the sheet P from a sheet feed tray **181** via a first sheet conveyance path **K11**. Conveyance rollers convey the sheet P fed from the sheet feed tray **181** to the coating apparatus **150** as the sheet processing apparatus that performs the pre-processing via the first sheet conveyance path **K11**.

[0116] In the present modification, the sheet feeder **180** is configured to feed cut sheets stored in the sheet feed tray **181** but may be configured to feed roll sheets.

[0117] Subsequently, the sheet P conveyed to the coating apparatus **150** is conveyed to a coating apparatus main portion **151** via a second sheet conveyance path **K12**. The coating apparatus main portion **151** applies coating liquid as pre-processing liquid to the lower face of the sheet P that is a front face during image formation to reduce bleeding and bleed-through, which is a coating process. After the coating process, the sheet P coated with the coating liquid is conveyed to a fourth sheet conveyance path **K14** that is a reverse passage. After the sheet P enters the fourth sheet conveyance path **K14**, the conveyance direction is reversed, and the sheet is turned over. The sheet face coated with the coating liquid becomes the front face (an upper face). Subsequently, the sheet P is conveyed to the inkjet printer **99** as the image forming apparatus via a third sheet conveyance path **K13**.

[0118] When a duplex print mode to form images on both sides of the sheet P is selected in the image forming apparatus **1**, the coating liquid is applied to both sides of the sheet P. To apply the coating liquid to both sides of the sheet P, the sheet P having one side coated with the coating liquid is conveyed to the fourth sheet conveyance path **K14**. After the sheet P enters the fourth sheet conveyance path **K14**, the conveyance direction is reversed, and the sheet is turned over. Subsequently, the sheet P is conveyed to a fifth sheet conveyance path **K15** (a duplex print conveyance path) to convey the sheet P to the coating apparatus main portion **151** again. The coating apparatus main portion **151** applies the coating liquid to the other side of the sheet P, and the sheet P is conveyed to the inkjet printer **99** as the image forming apparatus via the third sheet conveyance path **K13**.

[0119] The configuration and operation of the coating apparatus main portion **151** in the coating apparatus **150** is described below in detail with reference to FIGS. **3A** to **8**.

[0120] The sheet P conveyed to the inkjet printer **99** as the image forming apparatus passes through a sixth sheet conveyance path **K16**. Subsequently, a conveyance drum **102** conveys the sheet P. While the conveyance drum **102** conveys the sheet P, the ink is ejected to the front face (the upper face) of the sheet P to form a desired image (an ink image). At this time, since the coating liquid is applied to the front face of the sheet P as the pre-processing, the bleeding and the bleed-through of the image are prevented.

[0121] The sheet P on which the image is formed is conveyed to the dryer **185** that is the sheet processing apparatus to perform the post-processing via a seventh sheet conveyance path **K17**.

[0122] The sheet P conveyed to the dryer **185** passes through an eighth sheet conveyance path **K18** and is conveyed to a dryer unit **186** to dry the image on the front side of the sheet P which is referred to as one side of the sheet P below. The sheet P bearing the image dried is conveyed to the sheet ejection device **190** via a ninth sheet conveyance path **K19**.

[0123] When the above-described duplex print mode is selected, an image is formed on the back side of the sheet P which is referred to as the other side of the sheet P. To form the image on the other side of the sheet P, the sheet P is conveyed to a tenth sheet conveyance path **K.sub.20** (that is a reverse passage) after the ink image on the one side is dried. After the sheet P enters the tenth sheet conveyance path **K.sub.20**, the conveyance direction is reversed, and the sheet P is turned over. Subsequently, the sheet P is conveyed to an eleventh sheet conveyance path **K21** and a twelfth conveyance path **K22** that are duplex print conveyance paths to convey the sheet P to the conveyance drum **102** in the inkjet printer **99** as the image forming apparatus again. A desired

image (an ink image) is formed on the other side of the sheet P on the conveyance drum **102**, and the sheet P is conveyed to the dryer **185** again via the seventh sheet conveyance path K**17**. The dryer unit **186** dries the image on the other side of the sheet P, and the sheet P is conveyed to the sheet ejection device **190** via the ninth sheet conveyance path K**19**.

[0124] The sheet P conveyed to the sheet ejection device **190** is stacked on the an output tray **191** after passing through a thirteenth sheet conveyance path K**23**.

[0125] In this way, a series of operations in the image forming system **200** is completed.

[0126] In the image forming system **200** according to the modification, the coating apparatus **150** and the dryer **185** as the sheet processing apparatus includes a sheet conveying device that conveys the sheet P in the sheet conveyance paths K**12** to K**15** and K**18** to K**21** (formed by the first guide and the second guide that face each other). The sheet conveying device such as the coating apparatus **150** or the dryer **185** may be configured in the same manner as the sheet conveying device **49** described above with reference to FIGS. 2A to 6B.

[0127] Further, the sheet conveying device installed in the sheet feeder **180**, the inkjet printer **99** as the image forming apparatus, or the sheet ejection device **190** may be configured in the same manner as the sheet conveying device **49** described above with reference to FIGS. 2A to 6B.

[0128] The above-described configuration easily determines the position of a feeler in a closed cover with respect to a photosensor even if the cover has a deformation.

[0129] As described above, the image forming apparatus **1** according to the present embodiment includes the body side guide **54** as the first guide disposed in the body of the image forming apparatus **1** and the cover side guide **55** as the second guide that faces the body side guide **54** to form the sheet conveyance path K**1**. In addition, the image forming apparatus **1** includes the cover **50** including the cover side guide **55**, and the cover **50** can expose a part of or an entire of the body side guide **54**. Further, the image forming apparatus **1** includes the detector **90** detecting whether the sheet P exists in and passes through the first sheet conveyance path K**1**. The detector **90** includes the transmission-type photosensor **91** as the photosensor disposed in the body of the image forming apparatus **1** and the feeler **92** that can move with respect to the cover side guide **55**. The feeler **92** is configured to come in and out of the detection region **91a** of the transmission-type photosensor **91** in conjunction with the passage of the sheet P when the cover **50** closes the body of the image forming apparatus **1**. In addition, the positioner **59** is disposed in the body of the image forming apparatus **1**.

[0130] The positioner **59** contacts the contacted portion **92d** of the feeler **92** and determines the position of the feeler **92** with respect to the transmission-type photosensor **91** when the sheet P is not in the first sheet conveyance path K**1** formed by the cover **50** closing the body of the image forming apparatus **1**.

[0131] As a result, even if the cover **50** is deformed, the position of the feeler **92** with respect to the transmission-type photosensor **91** is easily determined after the cover **50** is closed.

[0132] The above-described embodiment and modification of the present disclosure are applied to the sheet conveying device **49** disposed in the image forming apparatus **1** forming the color toner image. However, the present disclosure may be also applied to a sheet conveying device disposed in a monochrome image forming apparatus.

[0133] Further, the above-described embodiment and modification of the present disclosure are applied to the sheet conveying device **49** provided for the image forming apparatus **1** that employs electrophotography. However, the present disclosure is not limited to the above-described sheet conveying device. For example, the present disclosure is also applicable to a sheet conveying device provided for an image forming apparatus that employs an inkjet method or a stencil printing machine. The present disclosure may be also applicable to an automatic document feeder as the sheet conveying device conveying a document as the sheet.

[0134] In the present embodiment and modification, the present disclosure is applied to the sheet conveying device **49** (in the image forming apparatus **1**) including the cover **50** that includes the

duplex printing conveyance path K5. However, the present disclosure may be applied to the sheet conveying device (the image forming apparatus) including the cover that does not include the duplex printing conveyance path.

[0135] In the present embodiment, the two guides that are the body side guide 54 and the cover side guide 55 form the second sheet conveyance path K1. Opening the cover 50 separates the cover side guide 55 in the cover 50 from the body side guide 54 in the body of the image forming apparatus 1. Two guides forming the first sheet conveyance path K0 may be configured so that opening the cover 50 separates the guide in the cover 50 from the guide in the body of the image forming apparatus 1.

[0136] Any of the cases described above exhibits substantially the same advantages as the advantages of the present embodiment and modification.

[0137] The above-described embodiments are illustrative and do not limit this disclosure. Thus, numerous additional modifications and variations are possible in light of the above teachings. For example, elements at least one of features of different illustrative and exemplary embodiments herein may be combined with each other at least one of substituted for each other within the scope of this disclosure and appended claims. The number, position, and shape of the components described above are not limited to those embodiments described above. Desirable number, position, and shape can be determined to perform the present disclosure.

[0138] In the present description, the term “sheet” is defined as any conveyed member that can be conveyed in the sheet conveyance path, such as general paper, coated paper, label paper, overhead projector (OHP) transparency, or a film sheet.

[0139] Note that aspects of the present disclosure may be applicable to, for example, combinations of first to twelfth aspects as follows.

First Aspect

[0140] In a first aspect, a sheet conveying device includes a first guide, a second guide, an opening and closing portion such as the cover, a detector, and a positioner. The first guide is disposed in an opened and closed portion such as the body. The second guide faces the first guide to form a sheet conveyance path. The opening and closing portion can expose a part or an entire of the first guide. The detector can detect a sheet passing through the sheet conveyance path. The detector includes a photosensor and a feeler. The photosensor is disposed in the opening and closing portion. The feeler is disposed to be movable with respect to the second guide. Under a state in which the opening and closing portion closes the opened and closed portion, the feeler moves in conjunction with passage of the sheet in the sheet conveyance path to come in and out of a detection region of the photosensor. The positioner is disposed in the opened and closed portion. When the sheet is not in the sheet conveyance path under the state in which the opening and closing portion closes the opened and closed portion, the positioner contacts a contacted portion of the feeler to determine a position of the feeler with respect to the photosensor.

Second Aspect

[0141] In a second aspect, the photosensor and the positioner in the sheet conveying device according to the first aspect are disposed on the first guide.

Third Aspect

[0142] In a third aspect, the photosensor and the positioner in the sheet conveying device according to the first aspect or the second aspect are integrally formed.

Fourth Aspect

[0143] In a fourth aspect, a rotatable angle of the feeler under a state in which the opening and closing portion closes the opened and closed portion is smaller than a rotatable angle of the feeler under a state in which the opening and closing portion opens the opened and closed portion in the sheet conveying device according to any one of the first to third aspects.

Fifth Aspect

[0144] In a fifth aspect, the second guide in the sheet conveying device according to any one of the

first to fourth aspects includes a second positioner contacting a second contacting portion of the feeler to determine a position of the feeler with respect to the second guide under a state in which the opening and closing portion opens the opened and closed portion.

Sixth Aspect

[0145] In a sixth aspect, the feeler in the sheet conveying device according to any one of the first to fifth aspects includes a lever portion and a detected portion. The lever portion is disposed to be rotatable about a rotation shaft with respect to the second guide. The lever portion is disposed at a position away from the rotation shaft. The lever portion can contact the sheet passing through the sheet conveying path. The detected portion is disposed at a position away from the lever portion on an opposite side with respect to the rotation shaft. The detected portion can come in and out of the detection region. The contacted portion is disposed at a position further away from the rotation shaft with respect to the detected portion.

Seventh Aspect

[0146] In a seventh aspect, the second guide in the sheet conveying device according to any one of the first to sixth aspects includes a biasing member biasing the feeler to move the feeler toward the positioner so that the feeler contact the positioner.

Eighth Aspect

[0147] In an eighth aspect, the second guide in the sheet conveying device according to any one of the first to seventh aspects includes an engaging portion engaging with an engaged portion of the first guide to determine a position in a width direction with respect to the first guide under a state in which the opening and closing portion closes the opened and closed portion.

Ninth Aspect

[0148] In a ninth aspect, the sheet conveying device according to any one of the first to eighth aspects further includes a support shaft and a reinforcement. The opening and closing portion is configured to be rotatable about the support shaft together with the second guide with respect to the opened and closed portion. The reinforcement reinforces the strength of the opening and closing portion is disposed in a region closer to the support shaft than a position at which the feeler is disposed.

Tenth Aspect

[0149] In a tenth aspect, an image forming apparatus includes the sheet conveying device according to any one of the first to ninth aspects. The opened and closed portion is the body of the image forming apparatus including a body side guide as the first guide. The opening and closing portion is an opening and closing cover rotating about the support shaft together with a cover side guide as the second guide with respect to the body of the image forming apparatus.

Eleventh Aspect

[0150] In an eleventh aspect, the image forming apparatus according to the tenth aspect includes a fixing device that is exposed under a state in which the opening and closing cover is opened.

Twelfth Aspect

[0151] In a twelfth aspect, a sheet processing apparatus that performs post-processing or pre-processing on a sheet on which an image is formed by an image forming apparatus includes the sheet conveying device according to any one of the first to ninth aspects.

[0152] The above-described embodiments are illustrative and do not limit the present invention. Thus, numerous additional modifications and variations are possible in light of the above teachings. For example, elements and/or features of different illustrative embodiments may be combined with each other and/or substituted for each other within the scope of the present invention.

Claims

1. A sheet conveying device comprising: a body; a first guide in the body; a cover to open and close the body to expose at least a part of the first guide at an opened state of the cover; a second guide in

the cover, the second guide facing the first guide to form a sheet conveyance path at a closed state of the cover; a detector to detect a sheet passing through the sheet conveyance path, the detector including: a feeler in the cover and movable relative to the second guide, the feeler including: a detected portion movable in conjunction with the sheet passing through the sheet conveyance path; and a contacted portion attached to the detected portion; and a photosensor in the body, the photosensor having a detection region in which the detected portion entering the detection region is detected by the photosensor; and a positioner in the body to contact the contacted portion to position the detected portion in the detection region when the sheet is not in the sheet conveyance path.

2. The sheet conveying device according to claim 1, wherein the photosensor and the positioner are disposed at the first guide.

3. The sheet conveying device according to claim 1, wherein the photosensor and the positioner are formed as a single component.

4. The sheet conveying device according to claim 1, wherein a rotatable angle of the feeler at the closed state of the cover is smaller than a rotatable angle of the feeler at the opened state of the cover.

5. The sheet conveying device according to claim 1, wherein the feeler includes another contacted portion, and the second guide includes another positioner contacting said another contacted portion to position the feeler with respect to the second guide at the opened state of the cover.

6. The sheet conveying device according to claim 1, wherein the feeler includes: a rotation shaft connected to the detected portion via another contacted portion; and a lever projecting from the rotation shaft toward the sheet conveyance path to contact the sheet passing through the sheet conveyance path, the detected portion and the lever are rotatable about the rotation shaft, the detected portion is separated from the lever and the rotation shaft by said another contacted portion, and the contacted portion is farther from the rotation shaft than the detected portion.

7. The sheet conveying device according to claim 1, further comprising a spring in the second guide to bias the feeler against the positioner.

8. The sheet conveying device according to claim 1, wherein the first guide has an engaged portion, and the second guide has an engaging portion engaging with the engaged portion, and the engaging portion positions the second guide with respect to the first guide in a direction: orthogonal to a direction in which the sheet passes through the sheet conveyance path; and parallel to a surface of the sheet.

9. The sheet conveying device according to claim 1, further comprising a support shaft disposed in the body and rotatably supporting the cover; and a reinforcement in the cover, wherein the reinforcement is closer to the support shaft than to the feeler.

10. An image forming apparatus comprising the sheet conveying device according to claim 1.

11. The image forming apparatus according to claim 10, further comprising a fixing device exposed by opening the cover.

12. A sheet processing apparatus comprising the sheet conveying device according to claim 1.
